

1. `python3 Linear_Classifiers.py --data simple --algorithm logistic`

```
nicholasmccracken@DESKTOP-H28MUAO HW04_Programming_3521 % python3
Linear_Classifiers.py --data simple --algorithm logistic
number of training data instances: (3, 8)
number of test data instances: (3, 2)
number of training data labels: (8, 1)
number of test data labels: (2, 1)
Accuracy: training set: 1.0
Accuracy: test set: 1.0
```

2. `python3 Linear_Classifiers.py --data starplus --algorithm logistic`

```
nicholasmccracken@DESKTOP-H28MUAO HW04_Programming_3521 % python3
Linear_Classifiers.py --data starplus --algorithm logistic
number of training data instances: (258391, 40)
number of test data instances: (258391, 14)
number of training data labels: (40, 1)
number of test data labels: (14, 1)
Accuracy: training set: 1.0
Accuracy: test set: 0.7142857142857143
```

3. `python3 Linear_Classifiers.py --data simple --algorithm perceptron`

```
nicholasmccracken@DESKTOP-H28MUAO HW04_Programming_3521 % python3
Linear_Classifiers.py --data simple --algorithm perceptron

number of training data instances: (3, 8)
number of test data instances: (3, 2)
number of training data labels: (8, 1)
number of test data labels: (2, 1)
Accuracy: training set: 1.0
Accuracy: test set: 1.0
```

4. `python3 Linear_Classifiers.py --data starplus --algorithm perceptron`

```
nicholasmccracken@DESKTOP-H28MUAO HW04_Programming_3521 % python3
Linear_Classifiers.py --data starplus --algorithm perceptron
number of training data instances: (258391, 40)
number of test data instances: (258391, 14)
number of training data labels: (40, 1)
number of test data labels: (14, 1)
Accuracy: training set: 1.0
Accuracy: test set: 0.7857142857142857
```